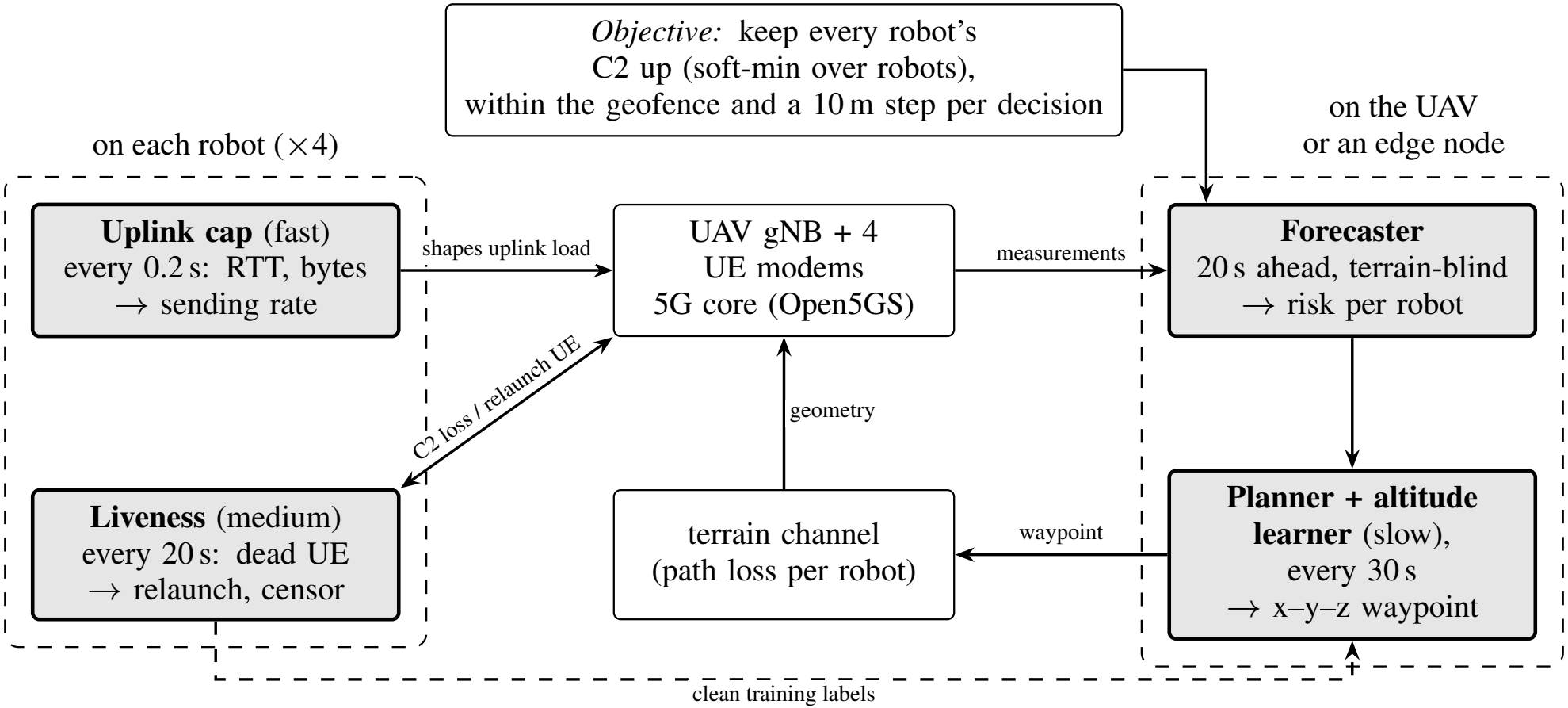
Objective: keep every robot's C2 up (soft-min over robots), within the geofence and a 10 m step per decision
on each robot (×4)
on the UAV or an edge node
Uplink cap (fast) every 0.2 s: RTT, bytes → sending rate
shapes uplink load
UAV gNB + 4 UE modems 5G core (Open5GS)
measurements
Forecaster 20 s ahead, terrain-blind → risk per robot
C2 loss / relaunch UE
geometry
Liveness (medium) every 20 s: dead UE → relaunch, censor
terrain channel (path loss per robot)
waypoint
Planner + altitude learner (slow), every 30 s → x–y–z waypoint
clean training labels